

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fourth Semester B.E. (E&C) (Credit System) Degree Examinations****Supplementary Examination - May - June 2023****20EC404 - DIGITAL SIGNAL PROCESSING**

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) Assume the missing data.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) State and prove circular time shift property of DFT.	8	L*2	1	1
	b) Compute circular convolution of $x_1 = \{2, 3, 1, 1\}$ and $x_2 = \{1, 3, 5, 3\}$ using DFT-IDFT method.	6	L3	1	1
	c) Given a length 10 sequence $x(n) = (0, 3, 4, -1, 4, 2, 8, 9, 2, 3)$. Find $y(n) = x((n - 7))_{10}$ using a suitable property.	6	L3	1	1
2.	a) State and prove circular convolution property of DFT.	8	L2	1	1
	b) Given $X(k)$ is a 14-point DFT of a real sequence $x(n)$. The first 8 samples of $X(k)$ are given by $X(0) = 12, X(1) = -1 + j3, X(2) = 3 + j4, X(3) = 1 - j5, X(4) = -2 + j2, X(5) = 6 + j3, X(6) = -2 - j3$ and $X(7) = 10$. Find the remaining samples using appropriate property. Also evaluate $\sum_{n=0}^{13} x(n) ^2$.	6	L3	1	1
	c) Evaluate circular autocorrelation of the sequence $x(n) = (1, 1, 2, 1)$.	6	L3	1	1
Unit – II					
3.	a) Obtain the output sequence $y(n)$ for a long input sequence $x(n) = \{1, 2, 3, 3, 2, 1, -1, -2, -3, 5, 6, -1, 2, 0, 2, 1\}$ and the impulse response of the system $h(n) = \{3, 2, 1, 1\}$ with a block length of $N=9$. Consider a filtering technique which has aliasing effect.	10	L3	2	1
	b) Calculate IDFT of the sequence $X(k) = (4, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.414, 0, 1 + j2.414)$ using DIT-FFT algorithm.	10	L3	2	1
4.	a) Perform block convolution of the two sequences given $x(n) = (3, -1, 0, 1, 3, 2, 0, 1, 2, 1)$ and $h(n) = (1, 1, 1)$ with a block length of 6 using a method that avoids aliasing effect.	10	L3	2	1
	b) Compute the 8-point DFT of the sequence $x(n) = (1, 2, 3, 4, 5, 6, 7, 8)$ using DIF-FFT algorithm.	10	L3	2	1
Unit – III					
5.	a) Convert an analog filter $H_a(s) = \frac{4}{(s+2)(s+3)}$ into a digital IIR filter using impulse invariance method with $T=2$ sec.	8	L3	3	1
	b) A Butterworth LPF has a pass band attenuation of 2.5dB at 2500π rad/sec and stop band attenuation of 90dB at 4000π rad/sec. What should be the minimum order of the filter? What should be the transition width if the order is doubled by maintaining the same pass band specification? Comment on the results.	6	L3	3	1
	c) A filter is given by the difference equation $y(n) - \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-2)$. Draw the direct form I and direct form II realizations. Also obtain the transfer function for the same.	6	L3	3	1

P.T.O.

6. a) The normalized transfer function of a second order Butterworth filter is given by

$$H_2(s) = \frac{1}{s^2 + 1.414s + 1}$$

Convert the analog filter to a digital filter with cut-off frequency of 0.5π rad/sec using Bilinear transformation at $T=1$ sec.

- b) Draw the direct form I and direct form II for the difference equation $y(n) + y(n-1) - 4y(n-3) = x(n) + 3x(n-2)$. Also obtain the transfer function for the same.
- c) A Butterworth filter has the following specifications with PB attenuation as -1 dB and PB frequency as 4 rad/sec. SB attenuation is -20 dB with SB frequency as 8 rad/sec. Compute the order and cut-off frequency. Calculate the poles and plot them.

8	L3	3	1
6	L3	3	1
6	L3	3	1

Unit – IV

7. a) The desired frequency response of an LPF is

$$H_d(e^{j\omega}) = e^{-j3\omega}; -\frac{3\pi}{4} \leq \omega \leq \frac{3\pi}{4}$$

$$= 0; \frac{3\pi}{4} < |\omega| \leq \pi$$

Design it using Hamming window ($M=7$) and obtain the system function.

- b) An FIR filter is given by the difference equation as $y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$. Realize this using lattice structure.

10	L3	4	1
10	L3	4	1

8. a) Design a low pass filter using rectangular window sequence of length $M=11$ for the frequency response

$$H_d(e^{j\omega}) = 1; |\omega| \leq \frac{\pi}{2}$$

$$= 0; \frac{\pi}{2} < |\omega| \leq \pi$$

Obtain the impulse response and transfer function.

- b) Obtain the direct form realization of linear phase FIR filter for the system function

i) $H(z) = \frac{1}{2} + \frac{1}{7}z^{-1} + \frac{1}{6}z^{-2} + \frac{1}{6}z^{-3} + \frac{1}{7}z^{-4} + \frac{1}{2}z^{-5}$.

ii) $H(z) = \frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2} + \frac{1}{5}z^{-3} + \frac{1}{8}z^{-4} + \frac{1}{4}z^{-5} + \frac{1}{2}z^{-6}$

10	L3	4	1
10	L3	4	1

Unit – V

9. a) Explain Braun Multiplier for unsigned number multiplication and discuss the issues in multiplication.
- b) What is overflow and underflow in MAC operation? Explain how saturation logic is used to avoid this.
- c) Find the signed Q15 representation for the decimal number 0.560123.
10. a) With a neat block diagram, explain Arithmetic Logic Unit.
- b) With a neat diagram explain Barrel shifter.
- c) Find the decimal equivalent of floating point binary number 110000000.01000000. Use single precision format.

8	L2	5	1
8	L2	5	1
4	L3	5	1
8	L2	5	1
8	L2	5	1
4	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
